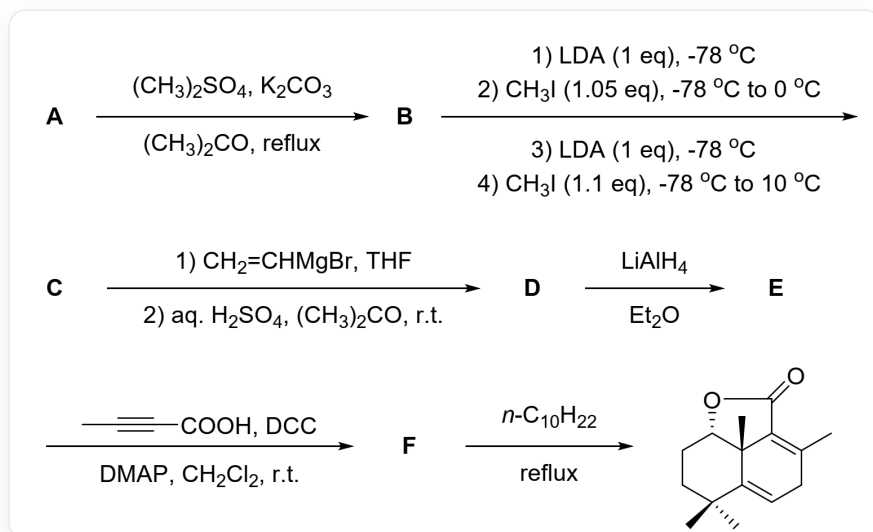


Question

The following reaction pathway exists:



The diagram shows a multi-step synthetic pathway, **A** generates **B** under $(\text{CH}_3)_2\text{SO}_4, \text{K}_2\text{CO}_3, (\text{CH}_3)_2\text{CO}$, *reflux* conditions, **B** successively undergoes:

(1) *LDA* (1 eq), -78°C . (2) CH_3I (1.05 eq), -78°C to 0°C . (3) *LDA* (1 eq), -78°C . (4) CH_3I (1.1 eq), -78°C to 10°C . to generate **C**, **C** successively undergoes: (1) CH_2CHMgBr , THF. (2) aq. H_2SO_4 , $(\text{CH}_3)_2\text{CO}$, r. t. to generate **D**,

D generates **E** under $\text{LiAlH}_4, \text{Et}_2\text{O}$ conditions, **E** generates **F** under CH_3CCCOOH , DCC, DMAP, CH_2Cl_2 conditions, **F** generates the final product under $n\text{-C}_{10}\text{H}_{22}$, *reflux* conditions, the final product is



Given that the chemical formula of **A** is $\text{C}_7\text{H}_{10}\text{O}_2$, if the substance exists in tautomeric forms, the most stable one is taken.

Then, which of the following statements is correct:

- A.** All other options are incorrect
- B.** **A** contains two carbonyl groups.
- C.** The mass fraction of oxygen in **B** is 20.75%.
- D.** The configuration of the chiral carbon in **C** is R and S.
- E.** The mass fraction of carbon in **D** is 80.44%.

F. Two exocyclic double bonds exist in **E**.

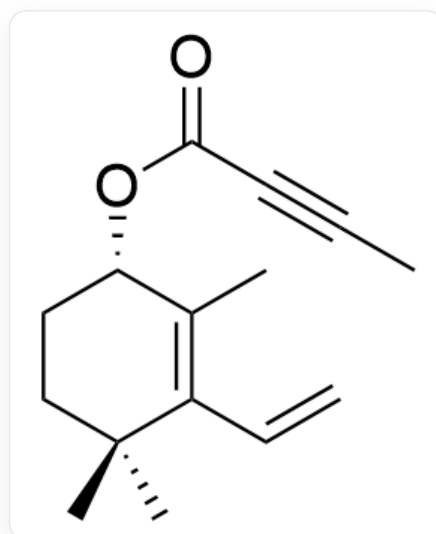
G. There exist two cycles in **F**.

Answer

Correct Answer: **E**

Detailed Explanation

Solving this problem through retrosynthetic analysis, we first observe that **E** to **F** uses an alkyne, and the final product only has double bonds and contains a six-membered ring with two double bonds. Combining with the conditions of the last step, it can be inferred that a D-A reaction occurred in the last step. Therefore, **F** is:



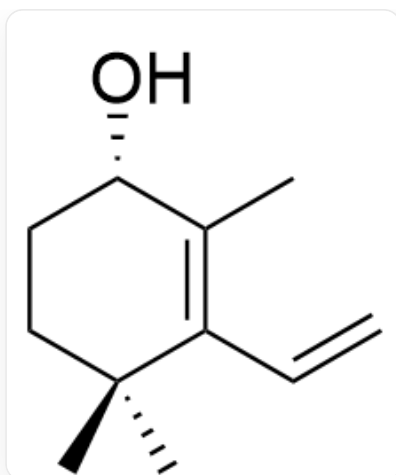
CC#CC(O[C@H]1CCC(C)(C(C=C)=C1C)C)=O

CHECKPOINT

1 PTS

F is CC#CC(O[C@H]1CCC(C)(C(C=C)=C1C)C)=O

The reaction conditions from **E** to **F** are typical substitution reaction conditions. Here, it is speculated that an esterification reaction occurs with the reactant. Therefore, **E** is:



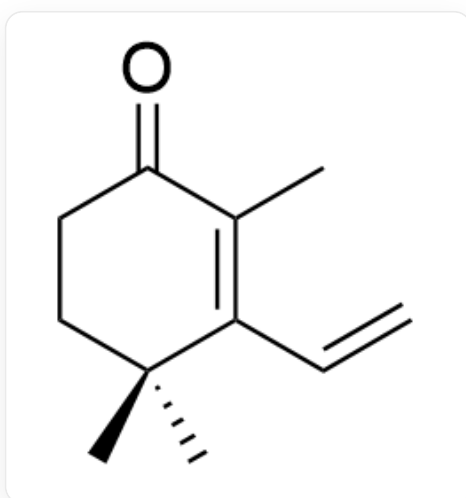
O[C@H]1CCC(C)(C(C=C)=C1C)C

CHECKPOINT

1 PTS

E is O[C@H]1CCC(C)(C(C=C)=C1C)C

D to **E** is under reduction reaction conditions, so it is speculated that this step is the reduction of the carbonyl group. Therefore, **D** is:



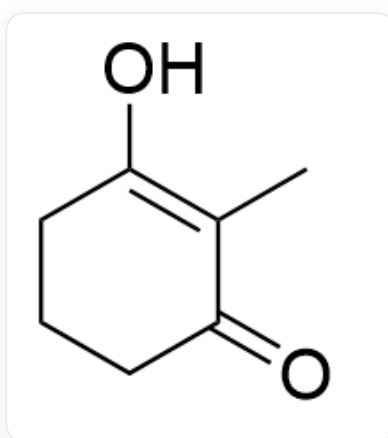
O=C1CCC(C)(C(C=C)=C1C)C

CHECKPOINT

1 PTS

D is O=C1CCC(C)(C(C=C)=C1C)C

At this point, it is more difficult to perform retrosynthesis. Therefore, we try to deduce the structure of **A** by combining the molecular formula. Observing that multiple methyl groups and one vinyl group are added during the process from **A** to **D**, it is speculated that **A** is a 1,3-diketone structure. Combining with the position of the methyl group in the molecule, it can be speculated that **A** is:



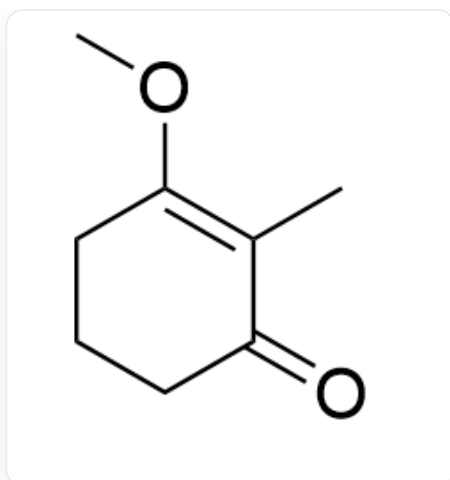
CC1=C(O)CCCC1=O

CHECKPOINT

1 PTS

A is CC1=C(O)CCCC1=O

A to **B** is the methylation of the hydroxyl group, so **B** is:



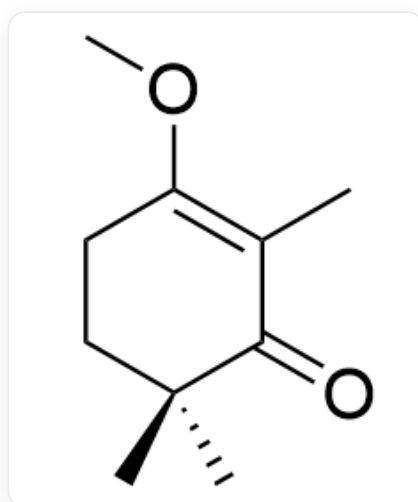
CC1=C(OC)CCCC1=O

CHECKPOINT

1 PTS

B is CC1=C(OC)CCCC1=O

Two reactions adding methyl groups are performed from **B** to **C**. Combining with the structure of **D**, it is easy to obtain that **C** is:



CC1=C(OC)CCC(C)(C)C1=O

CHECKPOINT

1 PTS

C is CC1=C(OC)CCC(C)(C)C1=O

After obtaining the structure, a simple calculation reveals that the mass fraction of oxygen in **B** is 22.83%, and the mass fraction of carbon in **D** is 80.44%. The remaining options can be judged as incorrect by direct observation.

Therefore, choose option E.